K.A.R.A FINANCE

Personal finance today. Business finance tomorrow.

PRODUCT BLUEPRINT

Feature architecture · Screen design · Technology stack · Development roadmap

Your money. Your decisions. Your future.

A proposed product blueprint for a colorful, cross-platform financial management app. The first release focuses on reliable personal finance tools, while the underlying architecture supports a future business workspace.

1. Product vision

KARA Finance is a personal financial operating system designed to help users understand their money, plan spending, build savings and make informed decisions. It can later expand into business finance without mixing personal and company records.

Product area	Purpose
Personal Finance	Income, expenses, accounts, budgets, savings goals, assets, liabilities and reports.
Business Finance	Separate business accounts, revenue, expenses, cash flow, invoices, customers and reports.
Intelligence	Financial insights, forecasting, receipt extraction and what-if simulations.

Core questions the app should answer

• How much money do I actually have?

• Where is my money going?

• How much can I safely spend?

• Am I making progress toward my goals?

• What could my financial position look like in the future?

Architecture principle

Keep personal and business workspaces separate, with separate ledgers, permissions and reports. A user's ownership of both does not make their transactions interchangeable.

2. Visual identity and design system

Use a colorful interface with a dark navy foundation, bright accents and clear cards. Default to a light dashboard with an optional dark mode. Color should communicate meaning rather than decorate every component.

Color	Hex	Meaning
Midnight	#17243D	Primary foundation and navigation
Violet	#7657F6	AI features and highlights
Mint	#35C7A0	Income, savings and positive cash flow
Blue	#3998F5	Accounts and information
Amber	#FFB84D	Goals, reminders and achievements
Coral	#F16F91	Expenses, losses and warnings
Cloud	#F3F5FA	Light surfaces and backgrounds

Design rules

• Use consistent spacing, rounded cards and readable typography.

• Keep the most important balance and monthly cash-flow figures near the top.

• Use charts only when they clarify a decision.

• Make key actions such as Add transaction easy to reach on both desktop and mobile.

• Support responsive layouts and accessible contrast.

3. Application navigation

Section	Main responsibility
Overview	At-a-glance financial dashboard and quick actions.
Transactions	Income, expenses, transfers, search and receipt attachments.
Accounts	Banks, cash, cards, savings and investments.
Budgets	Spending limits, progress and alerts.
Goals	Savings targets, milestones and contributions.
Analytics	Spending trends, income, savings rate and net worth.
AI Assistant	Questions, insights and scenario planning.
Business	Separate business workspace and operating finance.
Settings	Profile, currency, security, integrations and preferences.

Mobile navigation

Use a bottom navigation bar for Overview, Transactions, Goals, Analytics and More. Put less frequently used areas such as accounts, business, settings and reports inside More.

4. Screen-by-screen feature design

Screen 1 — Financial overview

The home screen should answer the most important questions within five seconds.

• Total available balance across selected accounts.

• Monthly income and expenses, with clear date range.

• Budget usage and remaining amount.

• Quick actions: Add, Transfer, Goal and Reports.

• Recent transactions and upcoming bills.

• Savings progress, account changes and weekly AI insights.

• Customizable dashboard widgets.

Screen 2 — Transactions

• Create income, expense or transfer records.

• Select account, payment method and category.

• Add merchant, date, notes, tags and receipt image.

• Search and filter by date, amount, merchant, category or account.

• Support recurring transactions and CSV import.

• Allow edits and corrections with a traceable history.

Screen 3 — Accounts and cards

• Support bank, cash, debit card, credit card, savings and investment accounts.

• Show current and available balances, transaction history, inflow and outflow.

• Allow multiple currencies and account reconciliation.

• Provide manual account setup before bank synchronization is available.

Screen 4 — Budget management

• Create weekly or monthly category budgets for rent, food, transport, shopping, education, entertainment and subscriptions.

• Show amount spent, remaining budget and days left.

• Offer threshold alerts, budget comparisons and optional rollover.

• Suggest budgets based on historical spending, with user approval.

Screen 5 — Savings goals

• Set a target amount, target date, current balance and priority.

• Calculate the required monthly and weekly contribution.

• Link a goal to a savings account without double-counting the money.

• Track emergency fund, education, travel, purchases and business capital.

Screen 6 — Analytics and reports

• Income versus expenses over time.

• Spending distribution by category.

• Monthly savings rate and budget performance.

• Account balance history and net worth progression.

• Recurring expense breakdown.

• Weekly, monthly, quarterly and yearly views; export reports.

Screen 7 — AI financial assistant

• Explain spending patterns and changes.

• Identify unusual transactions and recurring costs.

• Help build savings plans and compare scenarios.

• Answer questions using authorized, up-to-date app records.

• Require explicit confirmation before changing records or taking sensitive actions.

Screen 8 — Business finance

• Separate business dashboard, accounts and ledger.

• Track revenue, expenses, cash flow, profit and operating costs.

• Create and track invoices, customers and vendors.

- Organize payroll records, tax-related records and business goals.
- Generate monthly and annual business reports.

5. Intelligent and automation features

Feature	Expected behavior
Receipt scanner	Extract merchant, date, total and line items from a receipt image; prepare a draft transaction for confirmation.
Financial forecasting	Estimate future balances from history, expected income and scheduled bills; show assumptions and uncertainty.
Financial calendar	Display salary dates, rent, subscriptions, bills and planned savings contributions.
Financial health score	Summarize emergency savings, spending consistency, debt obligations, budget performance and savings progress.
Unusual spending detection	Flag transactions that differ materially from the user's normal patterns.
What-if simulator	Model purchases, moving costs, loans or business startup costs and show their effect on future cash flow.

AI boundaries

AI should explain and assist, not invent financial facts. All balances, transfers, totals and ledger updates must be computed by deterministic application logic. Forecasts should disclose their assumptions. Sensitive actions require explicit user confirmation.

6. Gamification and progression

Use a progression system to encourage consistent financial habits. Reward useful behaviors, not spending or simply opening the app.

• XP for recording expenses consistently.

• XP for staying within a budget.

• Milestones for reaching savings targets.

• Achievements for completing a monthly financial review.

• Streaks for healthy tracking habits, with no penalty for missing a day.

7. Technical architecture

Recommended stack

Layer	Technology	Role
Web frontend	Next.js + React	Responsive web application.
Mobile	React Native + Expo	iOS and Android application.
Styling	Tailwind CSS / shared design tokens	Consistent interface and components.
Charts	Recharts	Analytics visualizations.
Backend	Supabase	Backend services and database platform.
Database	PostgreSQL	Structured financial records.
Authentication	Supabase Auth	User sign-in and identity.

Layer	Technology	Role
Storage	Supabase Storage	Receipt images and exports.
AI	OpenAI API	Insights, extraction and conversational analysis.
Hosting	Vercel	Web deployment.
Version control	GitHub	Source control and collaboration.

High-level system flow

Web app Next.js / React	**Mobile app** React Native / Expo
Application API and business logic Authentication · Permissions · Validation · Financial calculations	
PostgreSQL database Financial records and ledgers	**AI services** Insights and predictions
Storage and integrations Receipts · Exports · Authorized financial data providers	

Build the web application first. Keep domain logic and financial calculations in a shared backend so the mobile app can use the same rules and data.

8. Database architecture

Suggested core entities

Table	Purpose
users	User profiles and identity references.
workspaces	Personal and business environments.
accounts	Bank, cash, card, savings and investment accounts.
transactions	User-facing transaction records.
ledger_entries	Debit and credit postings supporting accurate accounting.
categories	Income and expense classification.
budgets	Budget limits and periods.
goals	Savings goals and progress.
recurring_rules	Rules for scheduled transactions.
assets / liabilities	Net worth components, debts and obligations.

Table	Purpose
receipts	Receipt metadata and file references.
notifications	Alerts, reminders and delivery status.
businesses	Business profiles and ownership relationships.
invoices	Invoice headers, line items and payment status.
customers / vendors	Business counterparties.

Accounting integrity rules

• Use a double-entry ledger for financial postings.

• A transfer between two owned accounts changes each account balance but is not income or an expense.

• Store monetary amounts as integer minor units or fixed-precision decimals, never binary floating-point values.

• Store currency on every relevant account and transaction.

• Record timestamps consistently and display them in the user's local timezone.

• Keep an audit trail for corrections and important financial changes.

• Use database transactions to ensure related ledger postings succeed or fail together.

9. Security, privacy and South Korea considerations

• Use strong authentication and optional two-factor authentication.

• Enforce row-level security and least-privilege access in the database.

• Encrypt data in transit and at rest; protect secrets and API keys server-side.

• Do not store bank passwords, card PINs or card security codes.

• Require explicit consent for financial-data integrations.

• Maintain audit logs, backups, recovery procedures and account deletion/export tools.

• Support KRW as the default currency, with optional multi-currency handling.

• Provide Korean and English labels if useful.

• Investigate authorized Open Banking or financial data providers for Korean accounts; availability and approval requirements vary.

• Treat tax and accounting compliance as a separate product and legal workstream before marketing the app for formal compliance.

10. Development roadmap

Stage	Scope	Priority
1. Core finance	Authentication, personal dashboard, accounts, income/expense tracking, categories and transaction history.	Build first
2. Planning	Budgets, savings goals, recurring bills, financial calendar and notifications.	High
3. Intelligence	Analytics, reports, AI assistant, forecasting and what-if simulator.	After core data is reliable
4. Mobile	Mobile dashboard, receipt capture, push notifications, biometric sign-in and sync.	After web MVP
5. Business	Business workspaces, revenue, expenses, cash flow, invoices and reports.	Expansion
6. Automation	Authorized bank integrations, categorization, advanced monitoring and forecasting.	Later

Recommended first milestone (MVP)

A working web app with authentication, personal accounts, transaction entry, category management, an overview dashboard, budgets and savings goals. It should work well with manual entry before any bank connection or AI automation is added.

11. Suggested build order and acceptance criteria

Module	Definition of done
Accounts	Users can create accounts, set opening balances and reconcile balances.
Transactions	Users can add, edit and search income, expense and transfer records.
Ledger	Every posted transaction balances; transfers do not inflate income or expense totals.
Dashboard	Balances and monthly totals match the underlying ledger for the selected date range.
Budgets	Spending totals update correctly and alerts trigger at configured thresholds.
Goals	Contributions and progress are visible without double-counting account balances.
Security	Users cannot access another user's data; sensitive actions are protected and logged.

12. Product principles

- **Accuracy before decoration:** financial totals must be trustworthy.
- **Manual entry first:** the product remains useful without bank integrations.
- **AI with guardrails:** AI explains and suggests; deterministic code calculates and records.
- **Separate workspaces:** personal and business finances never silently mix.
- **Build in layers:** ship a useful personal finance MVP before expanding.

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This blueprint is a product and engineering starting point. It is not a substitute for professional accounting, tax, legal or financial advice. Validate financial integrations, privacy obligations and local compliance requirements before launch.